Samuel Sutherland

Computational physicist working on differentiable numerical methods and machine learning. PhD in differentiable simulation and gradient propagation through tensor network algorithms. Now working on ML research.

Education

UNSW — **Doctor of Physics**, *2021-2026*, *Thesis submitted, under examination*
- Primary Supervisor Prof. Michelle Simmons AC FRS

University of Oxford — **Master of Physics**, *2015-2019*, *1st class*
- Trained a model on satellite data to detect Pockets of Open Cells at unprecedented scale; resulting analysis won best paper at the ICML 2019 Climate Change AI workshop

Experience

Silicon Quantum Computing — **PhD Student**, *2021-2026*, University of New South Wales, Sydney
- Built multi-node DDP training with gradient-propagating collectives in PyTorch for Hamiltonian learning
- Derived and implemented a solver-agnostic backward pass for tensor network eigensolvers, eliminating the dependence of gradient cost and memory on solver iteration count
- Designed and tested a novel reservoir computing device, "Watermelon", based on an array of phosphorus-doped silicon quantum dots, which became a primary revenue stream for the company

Optiver — **Quantitative Researcher Intern**, *November 2023 - February 2024*, Sydney
- Evaluated a news-based trading signal on Bloomberg headlines; found no edge at actionable latency
- Developed improved estimators of intermediate-scale volatility in global equity indices

Ocado Technology — **ML Intern, then Innovation Engineer**, *Jul–Dec 2019, Apr–Dec 2020*, Hatfield
- Built an autonomous navigation stack on a mobile robot platform: SLAM-based mapping and routing with a vision model for obstacle detection, demonstrated end-to-end in a live office environment
- Applied self-supervised pretraining (MoCo, BYOL) to warehouse basket imagery for automated pick verification; found limited gains over supervised training given the abundance of implicit labels

Independent Research

Second-Order Jacobian Lens — **Interpretability study**, *2026*, samasutherland.github.io/projects
- Showed low-rank sketching of the Jacobian Lens with task-agnostic bases recovers no more than a random subspace, while a lens-derived basis reaches full performance at 10% of the dimensions
- Showed the second-order interaction of token-gated routing is step-like, so gradient-based estimation reports false negatives for most token pairs

little Language Models — **Transformer architecture experiments**, *2026*, samasutherland.github.io/llms
- Built a config-driven, Pydantic-validated framework that compares transformer variants at equal compute
- Made every result reproducible from a commit: containerised build, ephemeral GPU provisioning, and training through evaluation run automatically on push

Skills

Numerical — Tensor networks, differentiable programming, custom autograd, optimisation, quantum simulation

ML — Python, PyTorch, transformer architectures, self-supervised learning, distributed training

Selected Publications and Patents

Nature — M. B. Donnelly, Y. Chung, R. Garreis, ..., **S. Sutherland** *et al.*, "Large-scale analogue quantum simulation using atom dot arrays," (2026).

In Preparation — **S. A. Sutherland**, M. B. Donnelly, J. G. Keizer *et al.*, "A precision engineered Fermionic lattice for quantum-enhanced machine learning," (2026).

ACS Nano — A. D. Tranter, L. Kranz, **S. Sutherland** *et al.*, "Machine Learning-Assisted Precision Manufacturing of Atom Qubits in Silicon," vol. 18, no. 30, (2024).

WO/2024 — **S. Sutherland**, S. K. Gorman, C. Myers, and M. Y. Simmons, "Quantum machine learning devices and methods," *WO2024007054A1*, (2024).

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